

Programming Project 1: Heuristic Query Optimization

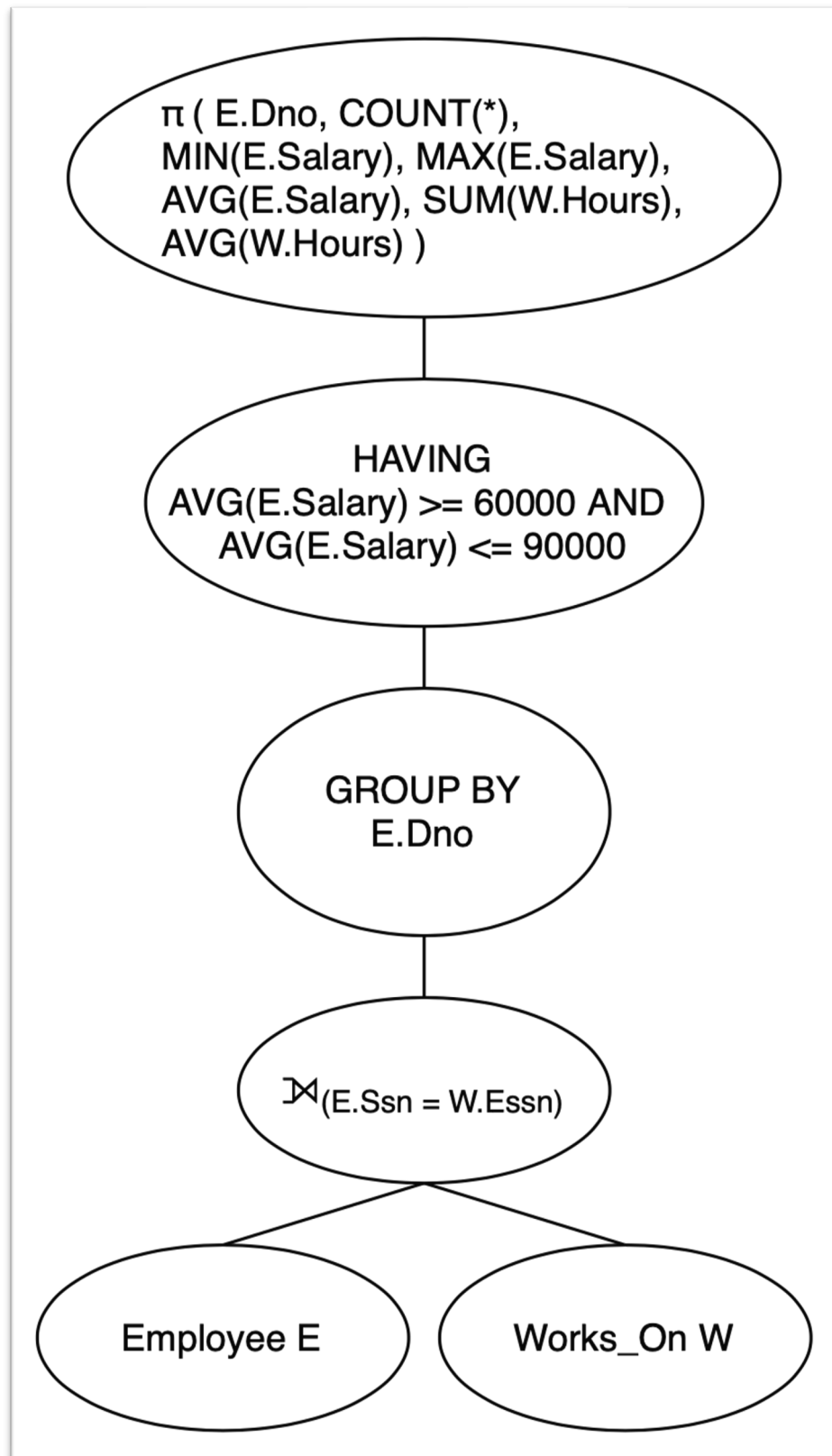
Example Inputs and Outputs

Example 3 (input3.txt)

Input SQL Query

```
SELECT
    E.Dno,
    COUNT(*),
    MIN(E.Salary),
    MAX(E.Salary),
    AVG(E.Salary),
    SUM(W.Hours),
    AVG(W.Hours)
FROM
    Employee E
    LEFT OUTER JOIN Works_On W ON E.Ssn = W.Essn
GROUP BY
    E.Dno
HAVING
    AVG(E.Salary) >= 60000 AND
    AVG(E.Salary) <= 90000;
```

Canonical Query Tree



Optimized Query Tree

Applied Rule #1 & #2: Identify conjunctive selections, break them up individually, and push them down as close as possible to the leaf nodes.

Not applicable.

Applied Rule #3: Apply Selections with Smallest Selectivity First.

Not applicable.

Applied Rule #4: Replace Cartesian Product + Selection with Join

Not applicable.

Applied Rule #5: Push Projections Down

